EPSR-test-ex38 makelatticeato: replicating an .ato box

Program(s): makelatticeato | chained passes: 1 | verification: ALL IDENTICAL (1 pass)

What this case tests

Exercises `makelatticeato` [1], which treats the contents of an existing `.ato` box as a unit cell and replicates it `na x nb x nc` times, optionally converting every atom into a tethered one-atom molecule — used to build larger or crystalline starting configurations for EPSR [2]. All arguments are tested on the command line, partly and fully on stdin.

Code exercised

Fortran (original, fortran_src/):

- `makelatticeato.f`: `makelatticeato` (replication, renumbering of bonds and rotational groups, conversion to tethered atoms)
- `ato_read_and_save.f`: `minitc` (read `.ato`), `moloutc` (write `.ato`)
- `file_and_directory_commands.f`: `search_for_files` (file prompt / directory listing), `get_full_filename`; `parse.f`: `parse`
- `molecom_mod.f90`: `testsize_labrot`, `testsize_rotgrps`; `reallocation_routines.f`: `reallocate1d_r`

Rust (port, rust/epsr-core/src/):

- `tools/makelatticeato.rs`: `run`
- `ato.rs`: `minitc`, `moloutc`
- `tools/setup_common.rs`: `search_for_files`, `get_full_filename`, `command_text`, `read_text`

Physical system

Molecules of the EPSR-test DMSO-water system [3] at 300 K. One dimethyl sulphoxide in a 21.544 Å box is replicated 2×2×2 (`dmso222.ato`: 8 molecules, cubic box 43.089 Å), 1×2×1 with conversion (`dmso121.ato`: 20 tethered atoms, non-cubic 21.54 × 43.09 × 21.54 Å) and 1×1×2 (`dmso112.ato`: 2 molecules, 21.54 × 21.54 × 43.09 Å). One water molecule (14.422 Å box) is replicated 3×2×1 with conversion (`waterlat.ato`: 18 tethered atoms, 43.27 × 28.84 × 14.42 Å).

Inputs

`DMSO.ato`, `WaterMolecule.ato` from EPSR-test. Commands: `makelatticeato DMSO 2 2 2 dmso222 n`; `makelatticeato WaterMolecule.ato 3 2 1 waterlat y` (conversion: `T00000` flags, new molecule types `moltype01..`); `makelatticeato` < `mla3.stdin` (everything asked, convert y); `makelatticeato DMSO.ato 1 1 2 dmso112` < `mla4.stdin` (only convert asked, n).

How the accuracy was verified

`commands.txt` (4 program calls, run once) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; no masking except where noted below). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in 1 pass.

Four program calls in one pass; compared outputs `dmso222.ato`, `waterlat.ato`, `dmso121.ato`, `dmso112.ato` and `dirlist.txt`, all identical. Both result layouts appear in the reference: the cubic header (8 molecules, side 0.43088699E+02) and the non-cubic header with three box lengths.

Notes

Fortran quirk reproduced: the convert answer is compared over the column range of the first word of the previous `parse`, because the Fortran parses an uninitialised variable; an answer `y` in column 1 is accepted.

References

[1] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).

[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).

[3] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).

[4] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.